Software Configuration Management

Outline

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What is SCM?

- Change management is commonly called *software configuration management* (SCM or CM)

- *Babich* defines Configuration Management as:
  - the art of identifying, organizing and controlling modifications to the software being built by a programming team.

- SCM activities have been developed to manage change throughout the life cycle of computer software

- SCM can be viewed as a quality assurance activity

What is SCM? (Contd.)

- SCM is a set of activities designed to manage change by:
  - Identifying s/w work products that are likely to change,
  - Establishing relationships among them,
  - Defining mechanisms for managing different versions of these work products
  - Controlling changes
  - And auditing and reporting on the changes made

Difference between Software Support and SCM

- Support is a set of s/w engg. activities that occur after software has been delivered to the customer and put into operation.

- While SCM is a set of tracking and control activities that initiate when the s/w engg project begins & end only when the s/w is taken out of operation

What is software configuration?

- The items that comprise all information produced as part of the software process are collectively called a ***software configuration***
- This information (also termed as output of software process) is broadly divided into:
  - Computer programs (both source level & executable forms)
  - Work products that describe the computer programs
  - Data (contained within program or external to it)

Sources of Changes

- **New business or market conditions**
  - dictate changes in product requirements or business rules.
- **New customer needs**
  - demand modification of data produced by information systems, functionality delivered by products, or services delivered by a computer-based system.
- **Reorganization or business growth/ downsizing**
  - causes changes in project priorities or software engg team structure.
- **Budgetary or scheduling constraints**
  - cause a redefinition of the system or product.

People involved in a typical SCM Scenario

- Project Manager
  - In charge of s/w group
  - Ensures that product is developed within time frame
  - Monitors development, recognizes & reacts to problems
    - by generating & analyzing reports about system status
    - By performing reviews
- Configuration Manager
  - In charge of CM procedures
  - Ensures that procedures & policies for creating, changing & testing of code are followed
  - To control changes in code, introduces mechanisms for official change requests, their evaluation & authoriza-tion.

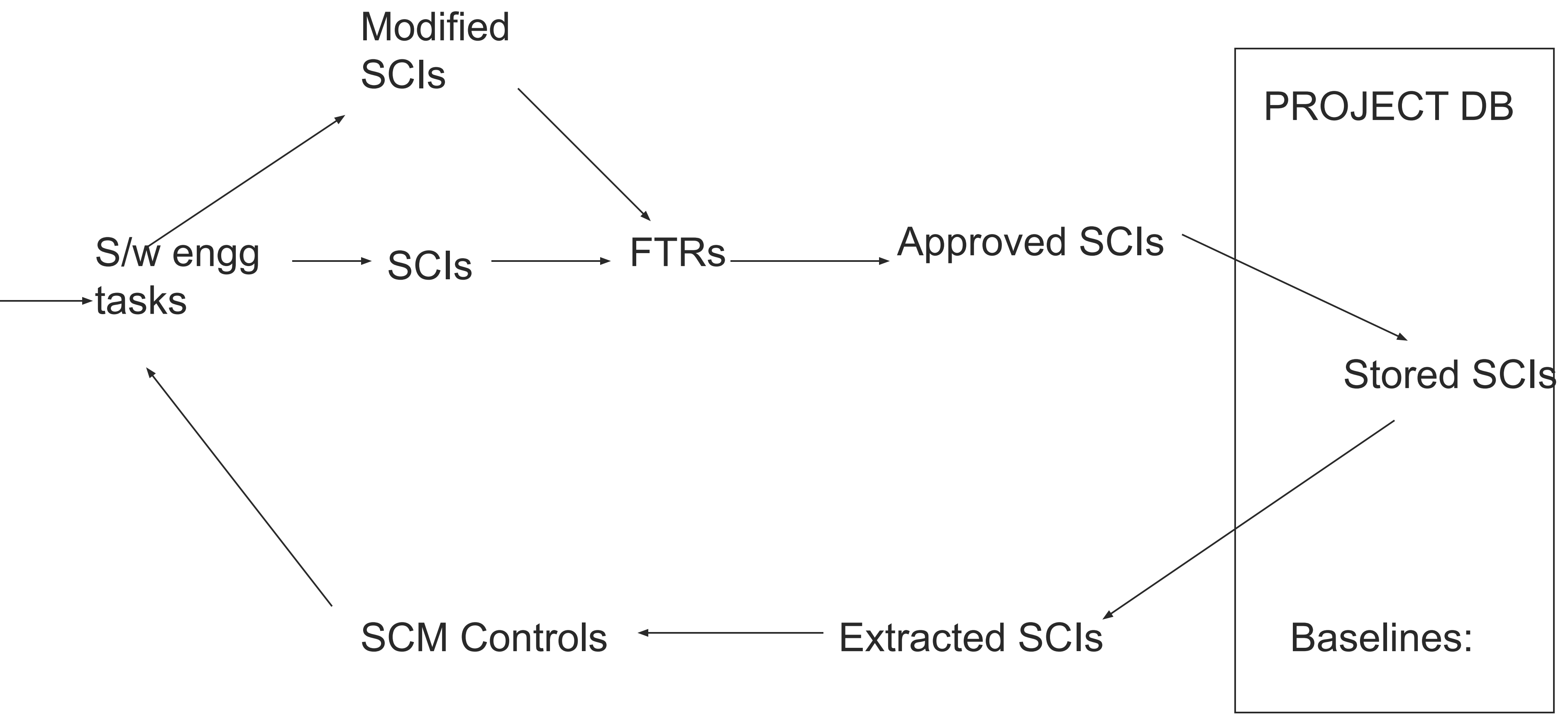
People involved in a typical SCM Scenario

- Software engineers
  - Work efficiently without interfering in each other's code creation, testing & supporting documentation.
  - But at the same time communicate & coordinate efficiently by
    - Updating each other about tasks required & tasks completed
    - Propagating changes across each others work by merging files
- Customers
  - Use the product
  - Follows formal procedures of requesting changes and for indicting bugs in the product

Baselines

- IEEE defines baseline as:
  - A specification or product that has been formally reviewed and agreed upon, that thereafter serves as basis for further development, and that can be changed only through formal change control procedures.
- Before a s/w configuration item becomes a baseline, changes may be made quickly & informally.
- However, once it becomes a baseline, formal procedures must be applied to evaluate & verify every change.
- Refer to figure 27.1 to see steps for forming a baseline.

Baselined SCIs and Project DB (fig. 27.1)



Configuration Objects

- SCIs are organized to form configuration objects.
- A config. object has a name, attributes and is related to other objects.
- Example:
  - Test Specification – Config. Object
  - It comprises of SCIs like Test Plan, Test Procedure, Test Cases
  - It is related to config. object SourceCode
- Config. Objects are related to other cong. objects, the relationship is either compositional (part-of relationship) or interrelated.
- In related config. objects, if one of them changes, the other one has to be changed.

SCM Repository

- In early days of s/w engg, software configuration items were maintained as paper documents (or punch cards), placed in files and folders.
- The problems of this approach were:
  - Finding a configuration item when it was needed was often difficult
  - Determining which items were changed, when and by whom was often challenging
  - Constructing a new version of existing program was error-prone & time consuming
  - Describing complex relationships among configuration items was virtually impossible
- Today SCIs are maintained in a project database or repository.
- It acts as the center for accumulation and storage of SCIs

Role of repository

- SCM Repository is a set of mechanisms & data structures that allow a software team to manage change in an effective manner.
- It provides functions of a DBMS but in addition has the following functions:
  - Data integrity
  - Info sharing
  - Tool integration
  - Data integration
  - Methodology enforcement
  - Document standardization

    - See page 778 for details of the above

SCM Process

- SCM process defines a series of tasks having 4 primary objectives
  - To identify all items that collectively define the s/w configuration.
  - To manage changes to one or more of these items.
  - To facilitate construction of different versions of an application.
  - To ensure that s/w quality is maintained as configuration evolves over time.

Version Control

- *Combines procedures and tools to manage the different versions of configuration objects created during the software process.*
- Version
  - A particular version is an instance of a system that differs in some way from other instances (difference may be enhanced performance, functionality or repaired faults).
  - A software component with a particular version is a collection of objects.
  - A new version is defined when major changes have been made to one or more objects
- Variant
  - A variant is a collection of objects at the same revision level and coexists with other variants.
  - Variants have minor differences among each other.

Version Control

- Object 1.0 undergoes revision
  - Object 1.0 -> Object 1.1
- If minor changes/corrections introduced to Object 1.1
  - Object 1.1.1 & 1.1.2 (variants)
- Functional changes, new modules added to Object 1.1 can lead to
  - Object 1.1 ->Object 1.2
- Major Technical Change
  - Object 1.0 ->Object 2.0

Version Control System

- A number of automated tools are available for version control.
- CVS (Concurrent version System) is an example which is based on RCS (Revision Control System)
- CVS features
  - Establishes a simple repository.
  - Maintains all versions of a file in a single named file by storing only the differences between progressive versions of the original file.
  - Protects against simultaneous changes to a file by establishing different directions for each developer

Change Control

- A Change request is submitted and evaluated to assess technical merit and impact on the other configuration objects and budget
- Change report contains the results of the evaluation
- Change control authority (CCA) makes the final decision on the status and priority of the change based on the change report
- Engineering change order (ECO) is generated for each change approved (describes change, lists the constraints, and criteria for review and audit)

Change Control (Contd.)

- Object to be changed is checked-out of the project database subject to access control parameters for the object
- Modified object is subjected to appropriate SQA and testing procedures
- Modified object is checked-in to the project database and version control mechanisms are used to create the next version of the software
- Synchronization control is used to ensure that parallel changes made by different people don't overwrite one another

Software Configuration Audit

- To ensure that a change has been properly implemented, FTRs and software config. audits are used.
- A Software Config. Audit complements the FTR by addressing the following questions:
  - Has the change specified by the ECO been made without modifications?
  - Has an FTR been conducted to assess technical correctness?
  - Was the software process followed and software engineering standards applied?
  - Do the attributes of the configuration object reflect the change?
  - Have the SCM standards for recording and reporting the change been followed?
  - Were all related SCI's properly updated?

Configuration Status Reporting

- CSR addresses the following questions:
  - What happened?
  - Who did it?
  - When did it happen?
  - What else will be affected by the change?
- CSR entries are made when:
  - A SCI is assigned new or updated identification
  - Change is approved by CCA
  - Config. Audit is conducted
- CSR is generated on regular basis.

THE END ….